7. Project Demonstration Phase

7.1 Introduction

The demonstration phase presents the completed EduGenie application and shows its major functionalities in a real-world usage scenario. It begins with registration and login and continues through tutoring, content generation, practice, planning, and progress tracking.

7.2 Objectives of Demonstration

- To demonstrate the overall functionality of EduGenie.
- To show the user authentication system.
- To demonstrate AI tutoring and concept explanation.
- To demonstrate notes summarization, quizzes, flashcards, and study plans.
- To display learning history and the progress dashboard.
- To show the interaction between frontend, backend, database, and Gemini.
- To present the final working application to the evaluator.

7.3 Demonstration Environment

The demonstration is performed in the configured development environment on a laptop with an internet connection, using VS Code to run the FastAPI server, a web browser for the frontend, the configured database, and a valid Google Gemini API key.

7.4 Demonstration Workflow

Step	Feature	Demonstration
1	Registration & Login	Create an account and log in securely.
2	AI Tutor	Ask a subject question and a follow-up question.
3	Concept Explanation	Explain one topic at beginner and advanced levels.
4	Notes Summarization	Paste notes and generate a summary with key points.
5	Quiz	Generate a quiz, answer it, and view the score and explanations.
6	Flashcards	Generate and review flashcards for a topic.
7	Study Plan	Enter subject, daily time, and exam date to get a plan.
8	History	Open previously saved chats, summaries, and quizzes.
9	Dashboard	View quiz scores and activity charts.
10	Logout	End the session.

7.5 Sample Scenario

A student preparing for an exam logs in and asks EduGenie to explain a topic in simple terms. The student pastes class notes to receive a summary, generates a ten-question quiz, reviews the explanations for wrong answers, creates flashcards for difficult terms, and requests a one-week study plan. The dashboard then shows the quiz score and topics studied.

7.6 Results and Observations

- The application meets the defined functional requirements.
- Gemini provides clear, relevant explanations and practice material.
- The interface is simple and easy to navigate.
- User data is protected through authentication.
- Saved history and the dashboard help learners track progress.

7.7 Limitations

- AI responses may occasionally be inaccurate and should be verified.
- An internet connection and Gemini API access are required.
- Response time depends on API availability and network speed.

7.8 Future Enhancements

- Voice input and spoken explanations.
- Upload and analysis of PDFs and images.
- Support for multiple languages.
- Mobile application.
- Adaptive learning paths based on quiz performance.
- Collaborative study groups and teacher dashboards.

7.9 Conclusion

EduGenie: Google Gemini Powered Learning Assistant combines conversational tutoring, content generation, practice, and progress tracking in a single application. The demonstration confirms that the system works as intended and provides learners with a convenient, personalized way to study.